AI Ethical Principles Case Study

NovaCare Health Systems

Responsible AI Framework for Healthcare

Course: Responsible Appl of AI (AIML-510-01R4A)
Assignment: 1.3 AI Ethical Principles Case Study
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Student Name	Student ID	Contribution
Vaibhavi Chirra	2926738	Researched and developed the fictional company's background, including its industry, size, operations, and need for a Responsible AI program.
Syeda Anees Fatima	2920992	Identified and explained the company's proposed AI use cases, including their business benefits and potential ethical risks.
Arshi Khan	2926773	Researched the AI ethical principles adopted by five major companies and gathered reliable sources for comparison.
Bhuvaneshwari Bibi Reddy	2935275	Created the comparison chart showing the common and unique AI ethical principles followed by the five companies.
Sugyeong Hong	2921747	Developed the recommended ethical principles for the fictional company and explained why common principles were included or excluded.
Tisungane Namacha	2921572	Drafted the company's AI ethics statement using guidance and best practices from Chapter 5 of *Ethical Machines*.
Laxmi Lahari Naidu Bugata	2930252	Reviewed the DoD Responsible AI Strategy and identified suitable methods and metrics for measuring AI ethics performance.
Akhila Chittimella	2925412	Prepared the conclusion, summarized the difficulties and questions raised by the workshop, and reviewed the entire presentation for formatting, consistency, citations, and accuracy.

NovaCare Health Systems

Fictional healthcare enterprise

Large U.S. healthcare network with 30 hospitals, 150+ outpatient facilities, and approximately 45,000 employees.

Serves about 6 million patients annually through hospitals, clinics, diagnostic centers, and telehealth services.

Operates enterprise EHR, cloud analytics, imaging, patient portals, and digital care platforms.

Strategic goal: use AI to improve clinical quality, access, operational efficiency, and patient experience.

Why Responsible AI matters: AI errors or bias can affect patient safety, privacy, treatment decisions, and trust.

AI Use Cases Under Consideration

AI-Assisted Medical Imaging

Detect abnormalities in X-rays, CT scans, and MRIs.
Risks: accuracy, bias, explainability, safety.

Patient Deterioration Prediction

Predict patients at risk of rapid clinical decline.
Risks: false alerts, data quality, unequal performance.

Generative AI Clinical Assistant

Summarize records and draft clinical documentation.
Risks: hallucinations, privacy, automation bias.

AI Scheduling & Resource Optimization

Forecast demand and optimize staffing and appointments.
Risks: employee fairness, transparency, over-automation.



Ethical AI Principles Across Major Organizations

Organization	Key Responsible / Ethical AI Principles
Microsoft	Fairness; reliability & safety; privacy & security; inclusiveness; transparency; accountability
Google	Social benefit; avoid unfair bias; safety; accountability; privacy; scientific excellence; appropriate use
IBM	Explainability; fairness; robustness; transparency; privacy
AWS	Fairness; explainability; privacy & security; safety; controllability; veracity & robustness; governance; transparency
U.S. DoD	Responsible; equitable; traceable; reliable; governable

Common themes: fairness, transparency, accountability, privacy, safety/reliability, and governance.

Recommended NovaCare Ethical Principles

1. Patient Safety & Reliability — Validate systems before deployment and monitor performance continuously.

2. Fairness & Equity — Test for unequal outcomes across relevant patient populations.

3. Privacy & Security — Protect health data throughout collection, training, deployment, and monitoring.

4. Transparency & Explainability — Provide meaningful information about capabilities, limitations, and consequential recommendations.

5. Human Oversight — Keep qualified professionals involved in high-impact clinical decisions.

6. Accountability — Assign clear human and organizational ownership for every AI system.

7. Governability — Maintain the ability to restrict, correct, suspend, or retire unsafe systems.

Excluded as standalone ethical principles: innovation and efficiency. They are business goals, but they must remain subordinate to safety, fairness, privacy, and accountability.

NovaCare Responsible AI Ethics Statement

NovaCare Health Systems is committed to developing, acquiring, and using artificial intelligence in ways that protect patients, support healthcare professionals, and promote trustworthy healthcare.

Our AI systems will be governed by patient safety and reliability, fairness and equity, privacy and security, transparency and explainability, meaningful human oversight, accountability, and governability.

AI will augment—not improperly replace—professional judgment in consequential clinical decisions. NovaCare will evaluate AI before deployment and continuously monitor it for safety, accuracy, bias, security, and unintended consequences. Systems that fail established ethical, clinical, or technical standards will be corrected, restricted, suspended, or retired.

The statement translates broad principles into clear commitments, oversight, monitoring, and corrective action.

Responsible AI Governance Model



Responsible AI Board: clinical leadership, AI/data science, cybersecurity, privacy/legal, compliance, patient representation, and ethics.

Required controls: AI impact assessment, data-quality and bias testing, privacy/security review, independent validation for high-risk systems.

Operational controls: documented limitations, human-oversight requirements, incident reporting, performance monitoring, periodic audits, and rollback/shutdown procedures.

Core idea: principles must be operationalized across the full AI lifecycle rather than treated as a one-time approval.

Conclusion and Workshop Reflection

Responsible AI requires more than accurate algorithms; it requires governance, testing, documentation, accountability, and continuous monitoring.

Across major frameworks, fairness, transparency, privacy, safety/reliability, accountability, and human control repeatedly appear as core themes.

Workshop difficulty/question: translating broad concepts such as fairness into measurable requirements can be challenging because different clinical contexts and patient populations may require different definitions of equitable performance.

Another question is how much explanation clinicians need from complex models before they can appropriately trust and act on recommendations.

Personal takeaway: effective Responsible AI requires collaboration among technical teams, clinicians, leaders, compliance professionals, and people affected by AI.

References

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Ethical Machines, Chapter 5. [Add the complete textbook citation from your course syllabus.]